SGN – Assignment #2

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Exercise 1: Uncertainty propagation

You are asked to analyze the state uncertainty evolution along a transfer trajectory in the Planar Bicircular Restricted Four-Body Problem, obtained as optimal solution of the problem stated in Section 3.1 (Topputo, 2013)*. The mean initial state $\mathbf{x}_i$ at initial time t_i with its associated covariance $\mathbf{P}_0$ and final time t_f for this optimal transfer are provided in Table 1.

1. Propagate the initial mean and covariance within a time grid of 5 equally spaced elements going from t_i to t_f, using both a Linearized Approach (LinCov) and the Unscented Transform (UT). We suggest to use $\alpha = 1$ and $\beta = 2$ for tuning the UT in this case. Plot the mean and the ellipses associated with the position elements of the covariances obtained with the two methods at the final time.

2. Perform the same uncertainty propagation process on the same time grid using a Monte Carlo (MC) simulation †. Compute the sample mean and sample covariance and compare them with the estimates obtained at Point 1). Provide the following outputs.

   - Plot of the propagated samples of the MC simulation, together with the mean and the covariance obtained with all methods in terms of ellipses associated with the position elements at the final time.
   - Plot of the time evolution (for the time grid previously defined) for all three approaches (MC, LinCov, and UT) of $3\sqrt{\max(\lambda_i(P_r))}$ and $3\sqrt{\max(\lambda_i(P_v))}$, where P_r and P_v are the 2x2 position and velocity covariance submatrices.
   - Plot resulting from the use of the MATLAB function `qqplot`, for each component of the previously generated MC samples at the final time.

   Compare the results, in terms of accuracy and precision, and discuss on the validity of the linear and Gaussian assumption for uncertainty propagation.

Table 1: Solution for an Earth-Moon transfer in the rotating frame.

Parameter	Value
Initial state $\mathbf{x}_i$	$\mathbf{r}_i$ = [-0.011965533749906, -0.017025663128129] $\mathbf{v}_i$ = [10.718855256727338, 0.116502348513671]
Initial time t_i	1.282800225339865
Final time t_f	9.595124551366348
Covariance $\mathbf{P}_0$	$\begin{bmatrix} +1.041e-15 & +6.026e-17 & +5.647e-16 & +4.577e-15 \\ +6.026e-17 & +4.287e-18 & +4.312e-17 & +1.855e-16 \\ +5.647e-16 & +4.312e-17 & +4.432e-16 & +1.455e-15 \\ +4.577e-15 & +1.855e-16 & +1.455e-15 & +2.822e-14 \end{bmatrix}$

*F. Topputo, "On optimal two-impulse Earth–Moon transfers in a four-body model", Celestial Mechanics and Dynamical Astronomy, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

†Use at least 1000 samples drawn from the initial covariance

1.1 Uncertainty propagation

1.1.1 Linearized approach

The first method used to propagate the mean and covariance is the Linearized Approach, commonly referred to as the LinCov method. This method focuses on studying displaced trajectories. To apply this approach, two key assumptions are made: first, that the uncertainties can be accurately described by Gaussian distributions; and second, that a linearized dynamic model is sufficiently accurate to describe the motion of the displaced trajectories. The general statement of the problem is presented in Equation 1, where $\mathbf{f}$ is the Equation 5 in Section 2.2 of (Topputo, 2013)$*$, and $\mathbf{\Phi}$ denotes the State Transition Matrix (STM).

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t) & \mathbf{x}(t_0) = \mathbf{x_0} \\ \dot{\mathbf{\Phi}} = \left[\dfrac{\delta \mathbf{f}}{\delta \mathbf{x}}\right] \mathbf{\Phi}(t_0, t) & \mathbf{\Phi}(t_0, t_0) = \mathbf{I}_{4x4} \end{cases} \tag{1}$$

Using the LinCov method, the propagated mean, as shown in (Equation 2), represents the mean of the final distribution. The propagated covariance is calculated using Equation 3.

$$\hat{\mathbf{x}}(t) = \mathbf{x}^*(t) \tag{2}$$

$$\mathbf{P}(t) = \mathbf{\Phi}(t_0, t)\mathbf{P_0}\mathbf{\Phi^T}(t_0, t) \tag{3}$$

The numerical continuation method has been utilized with this approach. For a time instant t_i, the propagated mean and covariance are those from the previous time step, t_{i-1}. It is important to note that for each propagation, the initial condition for the State Transition Matrix (STM) remains constant, $\mathbf{\Phi}(t_i, t_i) = \mathbf{I}_{4x4}$.

1.1.2 Unscented transform

The second method used for uncertainty propagation is the Unscented Transform (UT). Unlike methods that aim to represent the entire final probability density function (PDF) accurately, the UT focuses specifically on accurately computing the mean and covariance. This is accomplished by parameterizing the initial PDF using a small number of carefully chosen sample points, known as sigma points. The UT method assumes that the initial uncertainty is well-represented by a Gaussian distribution. The algorithm for the UT is as follows:

- computation $\mathbf{\Delta x}_i$, with (Equation 4), where the subscript stand for the $i-th$ column; $c = \alpha^2(N + k)$, with $\alpha = 1$, $\beta = 2$, $k = 0$;

$$\mathbf{\Delta x}_i = (\sqrt{c\mathbf{P}_x})_i \tag{4}$$

- creation of 9, (1 + 2N), sigma points using Eq.(Equation 5), where N is the dimension of the vector $\hat{\mathbf{x}}$;

$$\begin{cases} \boldsymbol{\chi}_0 = \hat{\mathbf{x}} \\ \boldsymbol{\chi} i = \hat{\mathbf{x}} + \mathbf{\Delta x}_i & i = 1, ..., N \\ \boldsymbol{\chi} i = \hat{\mathbf{x}} - \mathbf{\Delta x}_i & i = N+1, ..., 2N \end{cases} \tag{5}$$

- computation of the weights for mean and covariance, using (Equation 6);

$$\begin{cases} W_0^{(m)} = 1 - \dfrac{N}{c} \\ W_0^{(c)} = (2 - \alpha^2 + \beta) - \dfrac{N}{c} \\ W_i^{(m)} = W_i^{(c)} = \dfrac{1}{2c} \end{cases} \tag{6}$$

- propagation of the sigma points: $\boldsymbol{\Upsilon}_i = \mathbf{f}(\boldsymbol{\chi}_i) \qquad i = 0, ..., 2N$
- computation of weighted mean for the $i-th$ time instant, through (Equation 7);

$$\boldsymbol{\mu}_y = \sum_{i=0}^{2N} W_i^{(m)} \boldsymbol{\Upsilon}_i \tag{7}$$

- computation of weighted covariance for the $i-th$ time instant, through (Equation 8);

$$\boldsymbol{\sigma}_y = \sum_{i=0}^{2N} W_i^{(c)} \left(\boldsymbol{\Upsilon}_i - \boldsymbol{P}_y\right) \left(\boldsymbol{\Upsilon}_i - \boldsymbol{\mu}_y\right)^T \tag{8}$$

- repeat the last two steps iteratively to cover the entire time grid.

The Figure 1 shows that the mean and the covariance ellipse of the two methods are almost similar along x, while there is a noticeable difference along y.

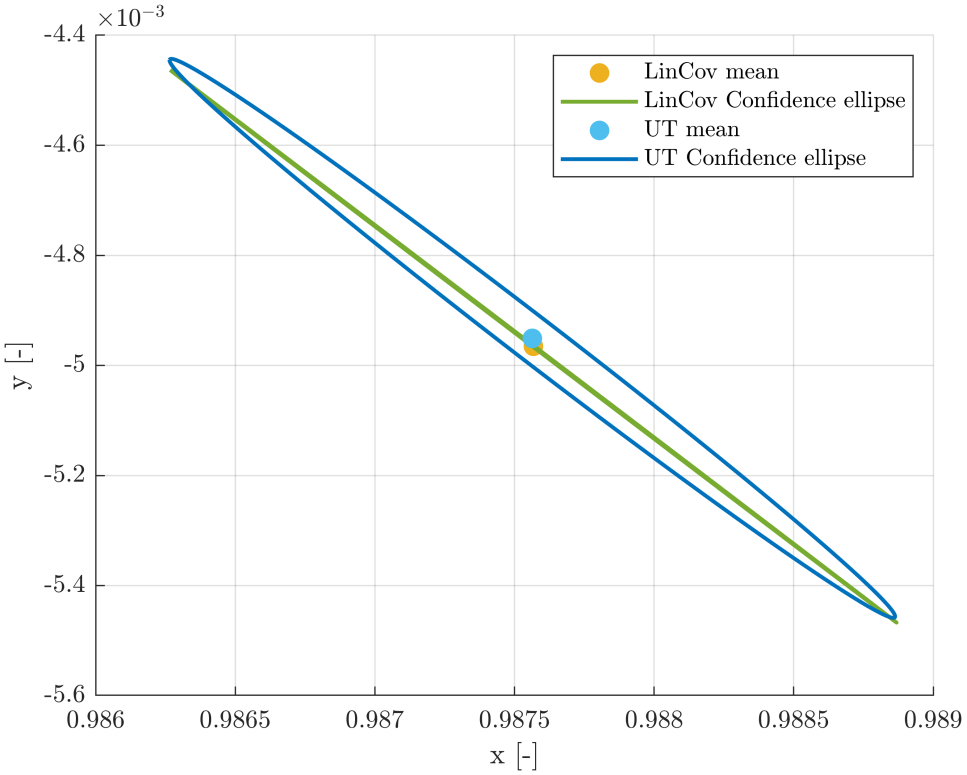


Figure 1: Confidence ellipses for LinCov and UT (@Earth Moon Barycenter adimensional rotating frame)

1.2 Monte Carlo simulation and results

1.2.1 Monte Carlo simulation

The Monte Carlo simulation is the most general method for propagating statistics. Given a random vector $\mathbf{x}$, with mean $\boldsymbol{\mu}_x$ and covariance $\mathbf{P}_x$, and a non-linear map $\mathbf{y} = \mathbf{g}(\mathbf{x})$, Monte Carlo simulation can model and analyze the statistical behavior of the system.
The algorithm is the following:

- generation of N_{MC} samples from $\boldsymbol{\mu}_x$ and $\mathbf{P}_x$;

- map all the samples using $\mathbf{g}(\mathbf{x})$;
- estimation of $\boldsymbol{\mu}_y$ and $\mathbf{P}_y$ using the sample mean and sample covariance, Equation 9 and Equation 10 respectively.

$$\boldsymbol{\mu}_y = \frac{1}{N_{MC}} \sum_{i=1}^{N_{MC}} \mathbf{y}_i \tag{9}$$

$$\boldsymbol{\sigma}_y = \frac{1}{N_{MC} - 1} \sum_{i=1}^{N_{MC}} (\mathbf{y}_i - \boldsymbol{\mu}_y)(\mathbf{y}_i - \boldsymbol{\mu}_y)^T \tag{10}$$

The samples were generated using the MATLAB function `mvnrnd(`$\boldsymbol{\mu}_x$`,` $\mathbf{P}_x$`,` N_{MC}`)` which returns a matrix of N_{MC} drawn from a multivariate normal distribution with mean vector $\boldsymbol{\mu}_x$ and covariance matrix $\mathbf{P}_x$. With this method, accuracy increases as N_{MC}, increases, but this also leads to a higher computational cost. For this simulation, a value of $N_{MC} = 1000$ was used.

1.2.2 Results

The first part of this section compares key quantities among the three propagation methods used: Linearized approach (linCov), Unscented Transform (UT), and Monte Carlo (MC).
From the Figure 2, it is observed that the mean computed with Monte Carlo is closer to the one obtained with UT, the same for the confidence ellipse which is similar to the UT one. From Figure 3 and Figure 4 is noticeable that the three approaches are almost similar, in fact,

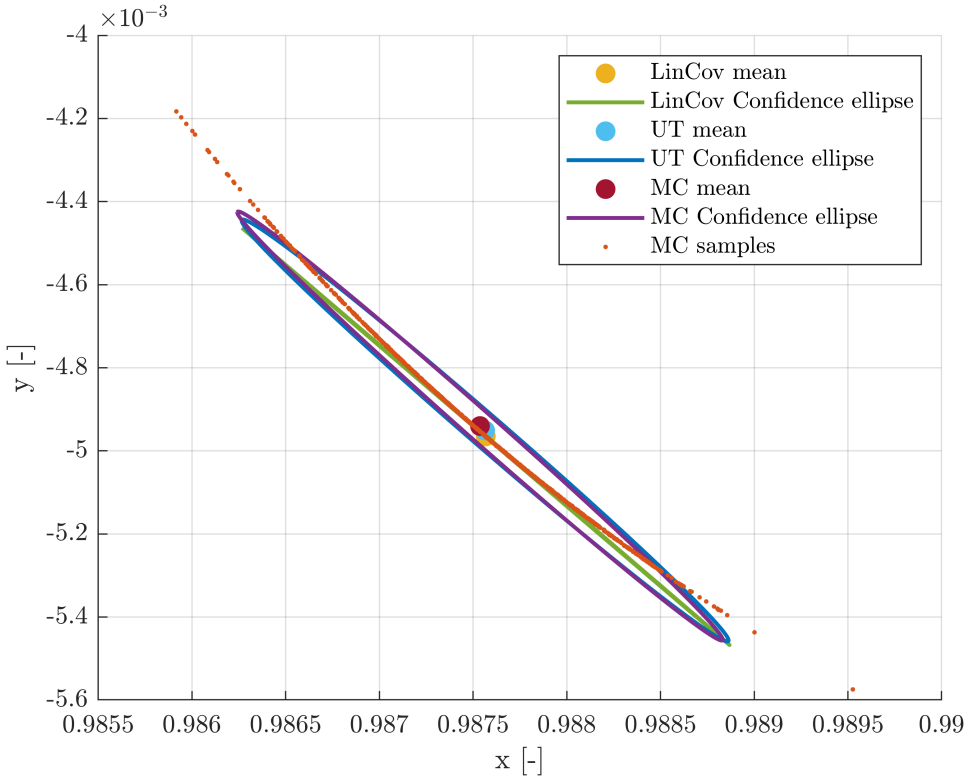


Figure 2: Confidence ellipses for LinCov, UT, and MC with propagated MC samples (@Earth Moon Barycenter dimensional rotating frame

the differences are indistinguishable. Based on the graphs, there are no significant differences apparent. This similarity is due to the fact that the eigenvalues of the covariance matrices at the final time are almost similar. Considering that by graphs is not possible to underline a quantitative difference, the Table 2 is retrieved.

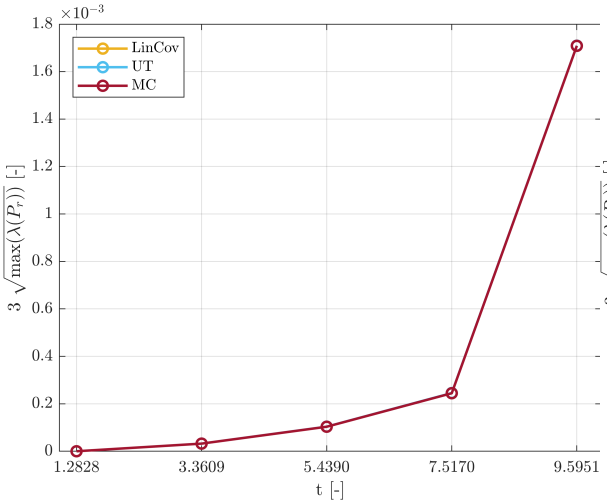


Figure 3: $3\sqrt{max(\lambda_i(\mathbf{P_r}))}$ time evolution

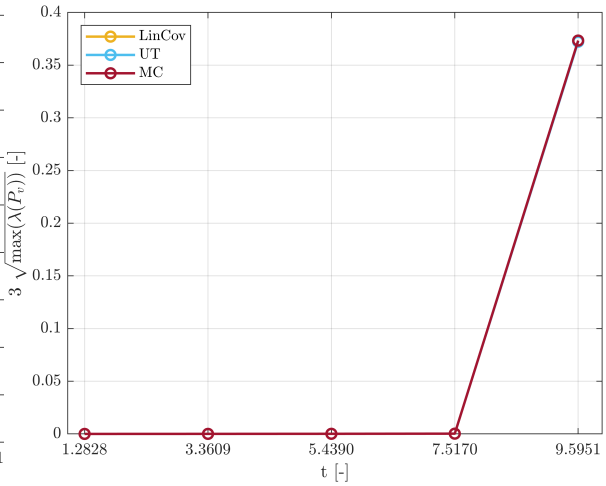


Figure 4: $3\sqrt{max(\lambda_i(\mathbf{P_v}))}$ time evolution

	LC - UT	MC - LC	MC - UT
Position [-]	3.735e-08	6.856e-07	6.482e-07
Velocity [-]	2.424e-04	7.934e-04	1.036e-03

Table 2: Difference of the last values of $3\sqrt{max(\lambda_i(\mathbf{P_r}))}$ and $3\sqrt{max(\lambda_i(\mathbf{P_v}))}$ between the three cases

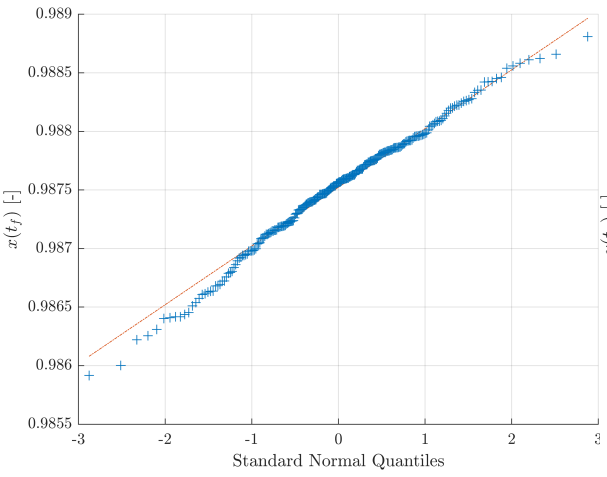


Figure 5: Q-Q Plot $x(t_f)$ of MC samples

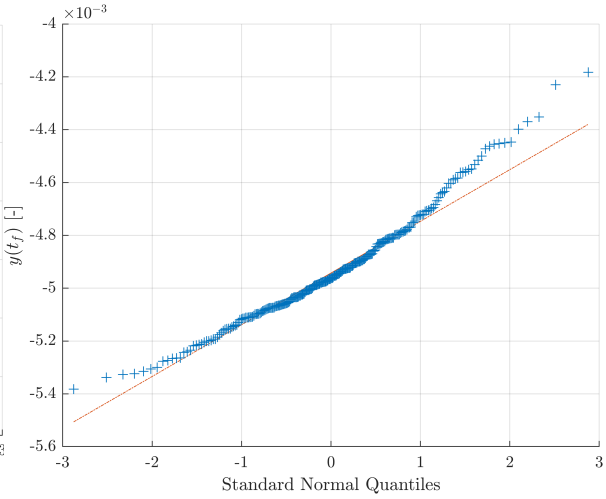


Figure 6: Q-Q Plot of $y(t_f)$ of MC samples

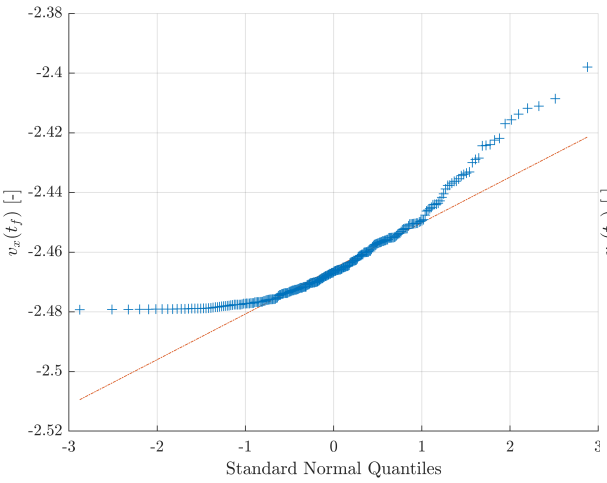


Figure 7: Q-Q Plot of $v_x(t_f)$ of MC samples

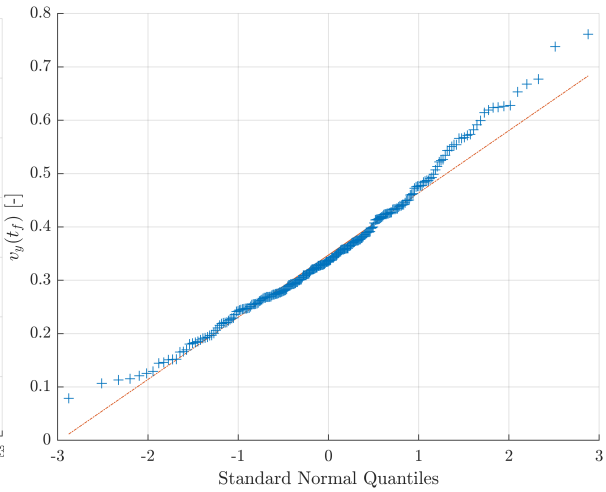


Figure 8: Q-Q Plot of $v_y(t_f)$ of MC samples

An important check can be done by comparing the distribution of each component of the previously generated MC samples at the final time against a standard normal distribution. By analyzing Figure 5, Figure 6, Figure 7, and Figure 8, can be concluded that:

- $\boldsymbol{x(t_f)}$, (Figure 5), the data closely follow the red line in the central portion, with slight deviations in the tails; the data are almost normal;
- $\boldsymbol{y(t_f)}$ (Figure 6), The points deviate significantly from the red line in both the lower tail and the upper tail; the overall shape suggests that the data might not be normally distributed;
- $\boldsymbol{v_x(t_f)}$ (Figure 7), the data do not follow the red line, so they are not normally distributed;
- $\boldsymbol{v_y(t_f)}$ (Figure 8), the data closely follow the red line across the range, with minimal deviations; the data are highly compatible with a normal distribution.

These conclusions lead that not all the components of the state can be described by a Gaussian distribution, so now it is clear why in Figure 1 and Figure 2 the confidence ellipse of the LinCov method is narrower compared to the ones of the UT and Monte Carlo.
This indicates that, within this scenario of relatively non-highly non-linear dynamics, both the Gaussian and linear assumptions are not valid.

Exercise 2: Batch filters

The Soil Moisture and Ocean Salinity (SMOS) mission, launched on 2 November 2009, is one of the European Space Agency's Earth Explorer missions, which form the science and research element of the Living Planet Programme.

You have been asked to track SMOS to improve the accuracy of its state estimate. To this aim, you shall schedule the observations from the three ground stations reported in Table 3.

1. *Compute visibility windows.* The Two-Line Elements (TLE) set of SMOS are reported in Table 4 (and in WeBeep as 36036.3le). Compute the osculating state from the TLE at the reference epoch t_{ref}, then propagate this state assuming Keplerian motion to predict the trajectory of the satellite and compute all the visibility time windows from the available stations in the time interval from $t_0 =$ `2024-11-18T20:30:00.000` (UTC) to $t_f =$ `2024-11-18T22:15:00.000` (UTC). Consider the different time grid for each station depending on the frequency of measurement acquisition. Report the resulting visibility windows and plot the predicted Azimuth and Elevation profiles within these time intervals.

2. *Simulate measurements.* Use `sgp4` and the provided TLE to simulate the measurements acquired by the sensor network in Table 3 by:

   (a) Computing the spacecraft position over the visibility windows identified in Point 1 and deriving the associated expected measurements.

   (b) Simulating the measurements by adding a random error to the expected measurements (assume a Gaussian model to generate the random error, with noise provided in Table 3). Discard any measurements (i.e., after applying the noise) that does not fulfill the visibility condition for the considered station.

3. *Solve the navigation problem.* Using the measurements simulated at the previous point:

   (a) Find the least squares (minimum variance) solution to the navigation problem without a priori information using
   - the epoch t_0 as reference epoch;
   - the reference state as the state derived from the TLE set in Table 4 at the reference epoch;
   - the simulated measurements obtained for the KOROU ground station only;
   - pure Keplerian motion to model the spacecraft dynamics.

   (b) Repeat step 3a by using all simulated measurements from the three ground stations.

   (c) Repeat step 3b by using a J2-perturbed motion to model the spacecraft dynamics.

   Provide the results in terms of navigation solution[‡], square root of the trace of the estimated covariance submatrix of the position elements, square root of the trace of the estimated covariance submatrix of the velocity elements. Finally, considering a linear mapping of the estimated covariance from Cartesian state to Keplerian elements, provide the standard deviation associated to the semimajor axis, and the standard deviation associated to the inclination. Elaborate on the results, comparing the different solutions.

4. *Trade-off analysis.* For specific mission requirements, you are constrained to get a navigation solution within the time interval reported in Point 1. Since the allocation of antenna time has a cost, you are asked to select the passes relying on a budget of 70.000 €. The cost per pass of each ground station is reported in Table 3. Considering this constraint,

[‡]Not just estimated state or covariance

and by using a J2-perturbed motion for your estimation operations, select the best combination of ground stations and passes to track SMOS in terms of resulting standard deviation on semimajor axis and inclination, and elaborate on the results.

5. *Long-term analysis.* Consider a nominal operations scenario (i.e., you are not constrained to provide a navigation solution within a limited amount of time). In this context, or for long-term planning in general, you could still acquire measurements from multiple locations but you are tasked to select a set of prime and backup ground stations. For planning purposes, it is important to have regular passes as this simplifies passes scheduling activities. Considering the need to have *reliable* orbit determination and *repeatable* passes, discuss your choices and compare them with the results of Point 4.

Table 3: Sensor network to track SMOS: list of stations, including their features.

Station name	KOUROU	TROLL	SVALBARD
Coordinates	LAT = 5.25144° LON = -52.80466° ALT = -14.67 m	LAT = -72.011977° LON = 2.536103° ALT = 1298 m	LAT = 78.229772° LON = 15.407786° ALT = 458 m
Type	Radar (monostatic)	Radar (monostatic)	Radar (monostatic)
Measurements type	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]
Measurements noise (diagonal noise matrix R)	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km
Minimum elevation	6 deg	0 deg	8 deg
Measurement frequency	60 s	30 s	60 s
Cost per pass	30.000 €	35.000 €	35.000 €

Table 4: TLE of SMOS.

```
1 36036U 09059A   24323.76060260  .00000600  00000-0  20543-3 0  9995
2 36036  98.4396 148.4689 0001262  95.1025 265.0307 14.39727995790658
```

2.1 Visibility windows

To compute all visibility windows of the SMOS satellite from the ground stations listed in (Table 3) over the time interval $[t_0, t_f]$, several steps need to be made.
First, extract the state from the TLE in Table 4 at a t_{ref}, then propapate it using a two-body propagator from t_{ref} until t_0. Once the state is propagated to t_0 it is possible to proceed to compute the visibility windows.
The function used to compute the visibility windows operates as follows:

- computation of the time grid using the measurement frequency Table 3;
- propagation of the SMOS state across the time grid, then for each time instant:
  - calls of `cspice_sxform` function, to get the rotation matrix from the ECI to station Topocentric reference frame at a specific time instant;
  - calls of `cspice_spkezr` function to get the position and velocity of the ground station in ECI at a specific time instant;
  - computation of the relative state between SMOS and ground station
  - rotation of the relative state from ECI to station Topocentric reference frame;
  - calls `cspice_reclat` function, which converts the rectangular coordinates into latitudinal ones, azimuth, elevation, and range;

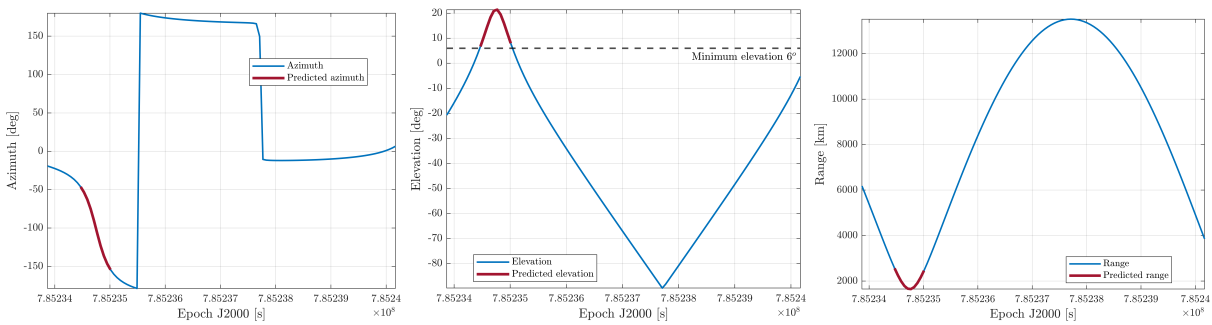
After the processing of all the states, the visibility windows are computed by checking which values of the elevation are above the minimum one requested by the ground stations.

Position[km]	Velocity [km/s]
-6065.41379482	0.60362446
3768.04689067	0.92674297
14.50092604	7.39076807

Table 5: SMOS state at t_{ref} (@ Earth ECI)

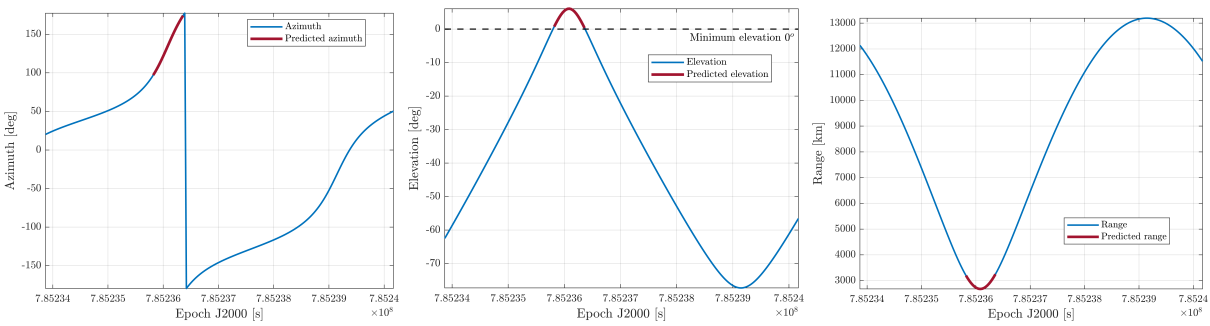
a [km]	e [-]	i [deg]	RAAN [deg]	ω [deg]	Mean Anomaly [deg]
7134.82931781	0.00124363	98.4342	148.4689	69.2929	290.8403

Table 6: SMOS osculating orbital elements at t_{ref} (@ Earth ECI)



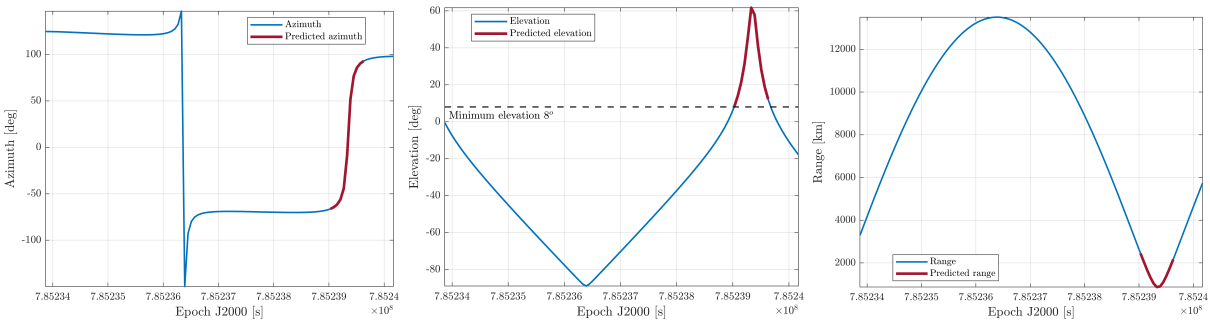
(a) Azimuth **(b)** Elevation **(c)** Range

Figure 9: SMOS coordinates for Kourou station(@Kourou Topocentric reference frame)



(a) Azimuth **(b)** Elevation **(c)** Range

Figure 10: SMOS coordinates for Troll station(@Troll Topocentric reference frame)



(a) Azimuth **(b)** Elevation **(c)** Range

Figure 11: SMOS coordinates for Svalbard station(@Svalbard Topocentric reference frame)

Station	Opening [UTC]	Closing [UTC]	Duration [s]	Number of points
KOUROU	2024-11-18T20:40:00.000	2024-11-18T20:49:00.000	540	10
TROLL	2024-11-18T21:02:30.000	2024-11-18T21:11:30.000	540	19
SVALBARD	2024-11-18T21:56:00.000	2024-11-18T22:06:00.000	600	11

Table 7: Time windows for each ground station

2.2 Measurements

The simulation of measurements involves computing the SMOS state from the TLE using the `sgp4` model. This model yields the satellite's state in the True Equator Mean Equinox (TEME) reference frame. To obtain measurements, it is necessary to convert this state to the Earth-Centered Inertial (ECI) reference frame. The function `teme2eci` is used for this conversion and requires nutation correction parameters for the date 2024-11-18, which are obtained from (*https://celestrak.org/SpaceData/EOP-All.txt*).
Measurements are then computed using the same procedure described for calculating visibility windows. Finally, random errors are introduced by applying `mvnrnd` with a covariance matrix as represented in Equation 11.

$$\mathbf{R} = \begin{bmatrix} \sigma^2_{Az,El} & 0 & 0 \\ 0 & \sigma^2_{Az,El} & 0 \\ 0 & 0 & \sigma^2_{Range} \end{bmatrix} \tag{11}$$

It is important to note that by adding errors, the Troll station lost a measurement. This issue arises primarily because the `sgp4` model accounts for perturbations due to the J_2 effect, whereas the time window calculations reported in Table 7 were performed using a purely Keplerian motion, the updated visibility windows are reported in Table 8.

Station	Opening [UTC]	Closing [UTC]	Duration [s]	Number of points
TROLL	2024-11-18T21:02:30.000	2024-11-18T21:11:00.000	510	18

Table 8: Updated Troll time window

2.3 Navigation problem

To address the navigation problem, a Batch filter has been implemented using the MATLAB function `lsqnonlin`. The goal of the filter is to find $\hat{\mathbf{x}}_\mathbf{k}$ that minimizes the weighted sum of the residuals, as described in Equation 12. Here, $\boldsymbol{\epsilon}$ represents the residuals, and $\mathbf{W}$ is the weight matrix, defined in Equation 13.

$$\mathbf{J} = \frac{1}{2}\boldsymbol{\epsilon}\mathbf{W}\boldsymbol{\epsilon}^{\boldsymbol{T}} \tag{12}$$

$$\mathbf{W} = \mathbf{R}^{-1} = \begin{bmatrix} \sigma_{Az,El}^{-2} & 0 & 0 \\ 0 & \sigma_{Az,El}^{-2} & 0 \\ 0 & 0 & \sigma_{Range}^{-2} \end{bmatrix} \tag{13}$$

The function passed to `lsqnonlin` propagates the initial guess of the state over the time grid. This propagation can either be purely Keplerian or account for the effects of acceleration due to J_2 perturbation. The implementation involves several steps: first, rotating the position from the ECI to the ECEF reference frame using `cspice_pxform` to obtain the rotation matrix; then, computing the acceleration due to J_2 using Equation 14; and finally, rotating the acceleration from ECEF back to ECI and adding it to the Keplerian acceleration.

$$\mathbf{a_{J2}} = \frac{3}{2}\mu J_2 \frac{\mathbf{r}}{r}\left(\frac{R_E}{r}\right)^2 \left(5\left(\frac{z}{r}\right)^2 - \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}\right) \tag{14}$$

For all states within the visibility windows computed earlier, measurements are obtained. These measurements are used to calculate residuals, which are then multiplied by the weight matrix to finalize the filter's objective.

2.3.1 Navigation Solution

The navigation problem is addressed using three distinct approaches:

- pure Keplerian motion with Kourou measurements only;
- pure Keplerian motion with all ground stations measurements;
- keplerian motion with J_2 effects and all ground stations measurements;

The results of these solutions are presented in Table 10. The nominal solution, detailed in Table 9, is obtained by propagating the SMOS TLE using `sgp4` at the reference epoch.t_0.
The standard deviation associated to the semimajor axis and inclination have been computed using a linear mapping of the estimated covariance from Cartesian state to Keplerian elements, Equation 15. $\mathbf{F}$ is the Jacobian of the transformation from Cartesian to Keplerian elements

derived with respect to the state components, evaluated at the navigation solution.

$$\mathbf{P}_{Kep} = \mathbf{F}\mathbf{P}_{Car}\mathbf{F}^T \tag{15}$$

Position[km]	Velocity [km/s]
3932.7688	4.8798
-1414.9188	-3.7632
5778.5030	-4.2329

Table 9: Nominal solution for SMOS navigational problem (@ Eartch ECI)

Stations	J2	Position [km]	Velocity [km/s]	σ_r [km]	σ_v [km/s]	σ_a [km]	σ_i [deg]
KOUROU	✗	3933.0857 -1416.0051 5780.0704	4.8808 -3.7616 -4.2374	9.8981	9.0828e-3	2.7976e+01	1.4671e-04
KOUROU TROLL SVALBARD	✗	3927.2728 -1411.3820 5779.8934	4.8882 -3.7649 -4.2337	1.0342	1.1147e-03	1.5096e-02	6.8607e-08
KOUROU TROLL SVALBARD	✓	3932.7678 -1414.9256 5778.4968	4.8798 -3.7632 -4.2329	2.0578e-02	2.0335e-05	2.4685e-05	6.2947e-11

Table 10: Solution for SMOS navigational problem (@ Eartch ECI)

The first solution, as might be expected, shows the greatest deviation because it is based on only 10 measurements from Kourou. The second solution, which incorporates measurements from all three stations, demonstrates a notable improvement, as it utilizes a total of 39 measurements. This increase in the number of measurements significantly enhances the accuracy of the state estimation.
Finally, the solution that most closely aligns with the nominal state of SMOS is the one that includes measurements from all three stations, while also accounting for the J_2 perturbation effect. This approach provides a more precise estimation by incorporating additional data and considering relevant perturbations, resulting in a solution that better reflects the true state of the satellite.

2.4 Trade-off analysis

Considering a real case scenario in which the allocation of each antenna has a cost, and considering the budget of 70.000€ an analysis of all the possible combinations of stations that respect the budget has been done. The propagation for all three cases has been performed by taking into account the J_2 perturbation.
By analyzing the results in Table 11 is possible to observe that all three combinations provide good results in terms of the standard deviation of inclination, while for which regards the standard deviation of the semimajor axis, the situation is different, in fact the results differs by orders of magnitude, the combination Kourou-Troll provides the worst solution, while the best result is given by the combination Kourou-Svalbard.

Stations	Total cost [€]	Position [km]	Velocity [km/s]	σ_a [km]	σ_i [deg]
KOUROU TROLL	65.000	3932.7944 -1414.9140 5778.4425	4.8798 -3.7632 -4.2328	4.3810e-02	5.6384e-09
KOUROU SVALBARD	65.000	3932.9184 -1414.9895 5778.4588	4.8797 -3.7632 -4.2329	7.0662e-05	4.1778e-09
SVALBARD TROLL	70.000	3932.9761 -1414.9007 5778.4162	4.8798 -3.7630 -4.2330	1.0936e-03	6.5853e-09

Table 11: Solution for SMOS navigational problem considering the cost per pass(@Eartch ECI)

2.5 Long-term analysis

In order to perform a long-term analysis is crucial to identify a sensible amount of time, which will help in the selection of a set of prime ground stations. Considering, that the orbital period of SMOS is approximately $T = 1.6660\,\text{h}$, so in a day it performs approximately 14 revolutions. By analyzing the elevation with the predicted one is possible to observe a repeatability of approximately 4 days for Kourou and 2 days for Troll and Svalbard, considering that an amount of time of 8 days has been selected.
Subsequently, the number of points in the visibility windows for each ground station has been computed, using the same procedure explained in subsection 2.1.
Figure 13, and Figure 14 shows how the Troll and Svalbard catch almost all the 14 revolutions in a day while Figure 12 shows how Kourou catches between 3 and 4 revolutions, this does not grant a reliable orbit determination.
Table 12 contains the number of points in all the visibility windows for all three ground stations, and it is possible to notice the big difference between Kourou and Troll or Svalbard.

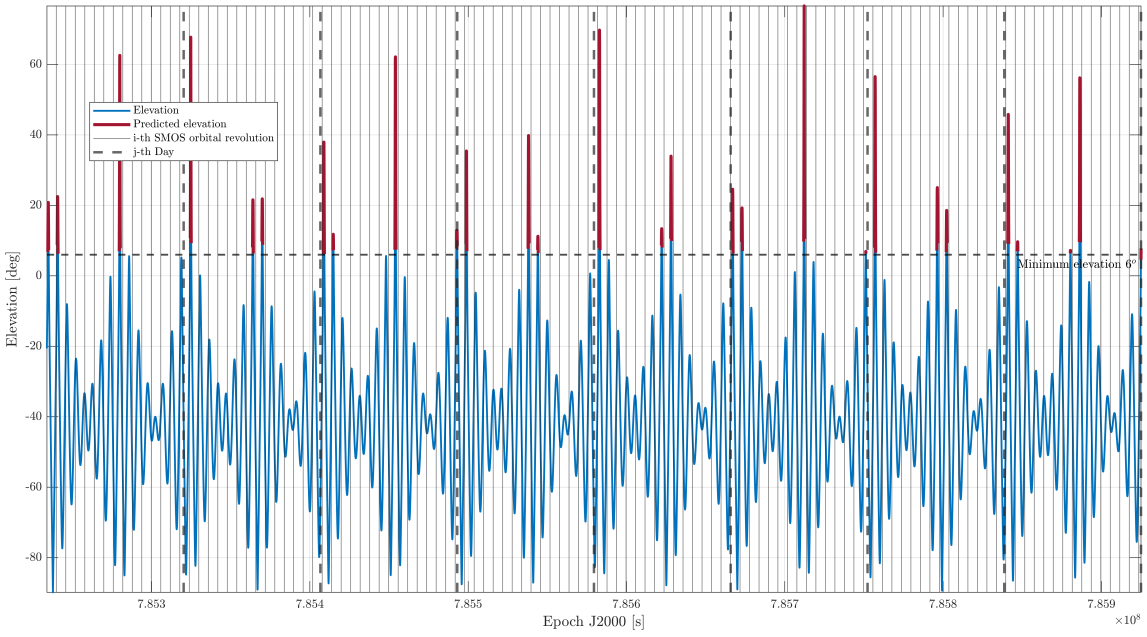


Figure 12: SMOS elevation for Kourou station(@Kourou Topocentric reference frame)

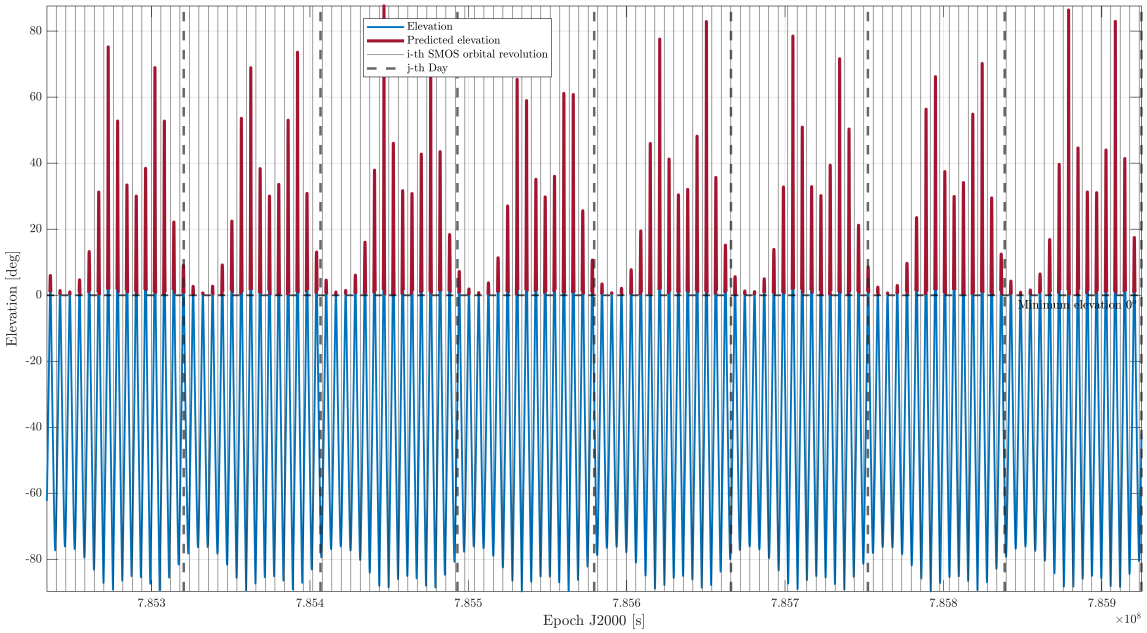


Figure 13: SMOS elevation for Troll station(@Troll Topocentric reference frame)

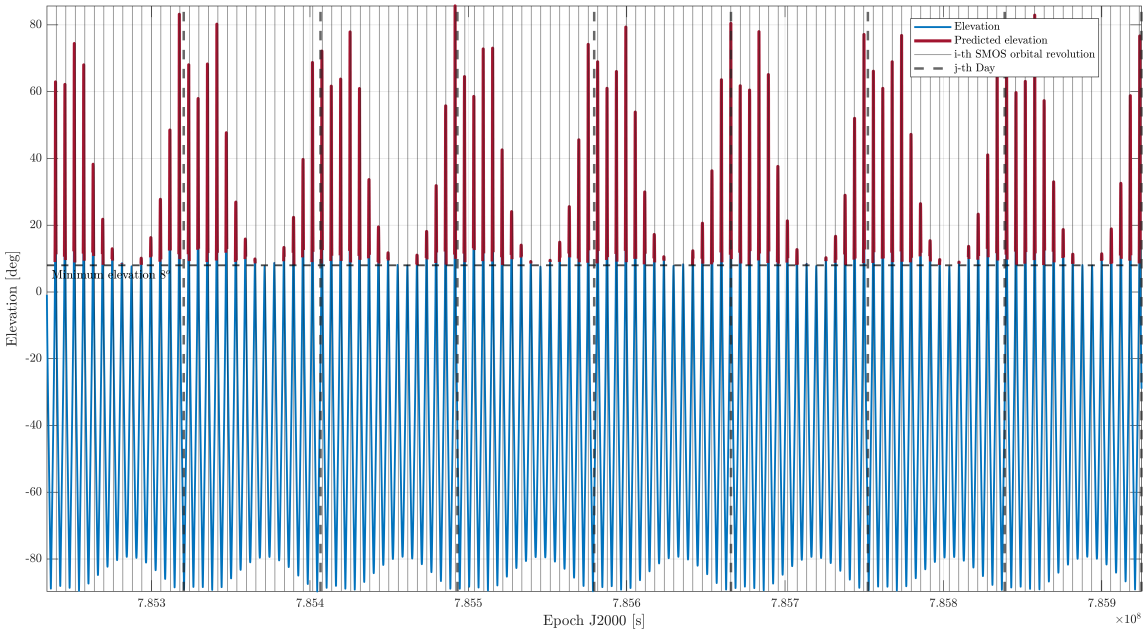


Figure 14: SMOS elevation for Kourou station(@Kourou Topocentric reference frame)

	KOUROU	TROLL	SVALBARD
N. of points	255	2776	949

Table 12: Number of points in the visibility windows for each ground station

The best solution in terms of prime ground station is Troll and Svalbard.
This conclusion differed a lot from the one obtained in the previous section in which the Kourou station has been chosen as one of the prime stations, this might have happened for the selected time interval, in which, as reported in Table 7, the number of points in the visibility window for Kourou station is equal to the Svalbard one.

Exercise 3: Sequential filters

An increasing number of lunar exploration missions will take place in the next years, many of them aiming at reaching the Moon's surface with landers. In order to ensure efficient navigation performance for these future missions, space agencies have plans to deploy lunar constellations capable of providing positioning measurements for satellites orbiting around the Moon.

Considering a lander on the surface of the Moon, you have been asked to improve the accuracy of the estimate of its latitude and longitude (considering a fixed zero altitude). To perform such operation you can rely on the use of a lunar orbiter, which uses its Inter-Satellite Link (ISL) to acquire range measurements with the lander while orbiting around the Moon. At the same time, assuming the availability of a Lunar Navigation Service, you are also receiving measurements of the lunar orbiter inertial position vector components, such that you can also estimate the spacecraft state within the same state estimation process.
To perform the requested tasks you can refer to the following points.

1. *Check the visibility window.* Considering the initial state $\mathbf{x}_0$ and the time interval with a time-step of 30 seconds from t_0 to t_f reported in Table 13, predict the trajectory of the satellite in an inertial Moon-centered reference frame assuming Keplerian motion. Use the estimated coordinates given in Table 14 to predict the state of the lunar lander. Finally, check that the lander and the orbiter are in relative visibility for the entire time interval.

2. *Simulate measurements.* Always assuming Keplerian motion to model the lunar orbiter dynamics around the Moon, compute the time evolution of its position vector in an inertial Moon-centered reference frame and the time evolution of the relative range between the satellite and the lunar lander. Finally, simulate the measurements by adding a random error to the spacecraft position vector and to the relative range. Assume a Gaussian model to generate the random error, with noise provided in Table 13 for both the relative range and the components of the position vector. Verify (graphically) that the applied noise level is within the desired boundary.

3. *Estimate the lunar orbiter absolute state.* As a first step, you are asked to develop a sequential filter to narrow down the uncertainty on the knowledge of the lunar orbiter absolute state vector. To this aim, you can exploit the measurements of the components of its position vector computed at the previous point. Using an Unscented Kalman Filter (UKF), provide an estimate of the spacecraft state (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. To initialize the filter in terms of initial covariance, you can refer to the first six elements of the initial covariance $\mathbf{P}_0$ reported in Table 13. For the initial state, you can perturb the actual initial state $\mathbf{x}_0$ by exploiting the MATLAB function `mvnrnd` and the previously mentioned initial covariance. We suggest to use $\alpha = 0.01$ and $\beta = 2$ for tuning the UT in this case. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both position and velocity.

4. *Estimate the lunar lander coordinates.* To fulfill the goal of your mission, you are asked to develop a sequential filter to narrow down the uncertainty on the knowledge of the lunar lander coordinates (considering a fixed zero altitude). To this aim, you can exploit the measurements of the components of the lunar orbiter position vector together with the measurements of the relative range between the orbiter and the lander computed at the previous point. Using an UKF, provide an estimate of the spacecraft state and the lunar lander coordinates (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. To initialize the filter in terms of initial covariance, you can refer to the initial covariance $\mathbf{P}_0$ reported in Table 13. For the initial state, you can perturb the actual initial state, composed by $\mathbf{x}_0$ and the latitude

and longitude given in Table 14, by exploiting the MATLAB function `mvnrnd` and the previously mentioned initial covariance. We suggest to use $\alpha = 0.01$ and $\beta = 2$ for tuning the UT in this case. Plot the time evolution of the error estimate together with the 3σ of the estimated covariance for both position and velocity.

Table 13: Initial conditions for the lunar orbiter.

Parameter	Value
Initial state $\mathbf{x}_0$ [km, km/s]	$\mathbf{r}_0$ = [4307.844185282820, -1317.980749248651, 2109.210101634011] $\mathbf{v}_0$ = [-0.110997301537882, -0.509392750828585, 0.815198807994189]
Initial time t_0 [UTC]	2024-11-18T16:30:00.000
Final time t_f [UTC]	2024-11-18T20:30:00.000
Measurements noise	σ_p = 100 m
Covariance $\mathbf{P}_0$ [km^2, km^2/s^2, rad^2]	`diag([10,1,1,0.001,0.001,0.001,0.00001,0.00001])`

Table 14: Lunar lander - initial guess coordinates and horizon mask

Lander name	MOONLANDER
Coordinates	LAT = 78° LON = 15° ALT = 0 m
Minimum elevation	0 deg

3.1 Visibility windows

In order to develop the filter is necessary to ensure that the orbiter and the lander are relative visibility for the entire time interval. Subsequently, the procedure adopted for computing the visibility window is the following:

- conversion of the lander coordinates, from geodetic to rectangular , using `cspice_georec`;
- propagation of the state of the orbiter (Table 13) from t_0 to t_f with a time-step of 30 seconds, then for each state in the time interval:
  - extraction of the rotation matrix with `cspice_sxform` from Moon-Centered inertial frame to Moon-Centered fixed frame;
  - rotation of the orbiter state into Moon-Centered fixed frame;
  - computation of the relative position between orbiter and lander;
  - extraction of the rotation matrix using `cspice_pxform` from Moon-Centered fixed frame to Moonlander Topocentricreference frame;
  - rotation of the relative position into Moonlander Topocentric reference frame;
  - computation of the elevation using `cspice_reclat`

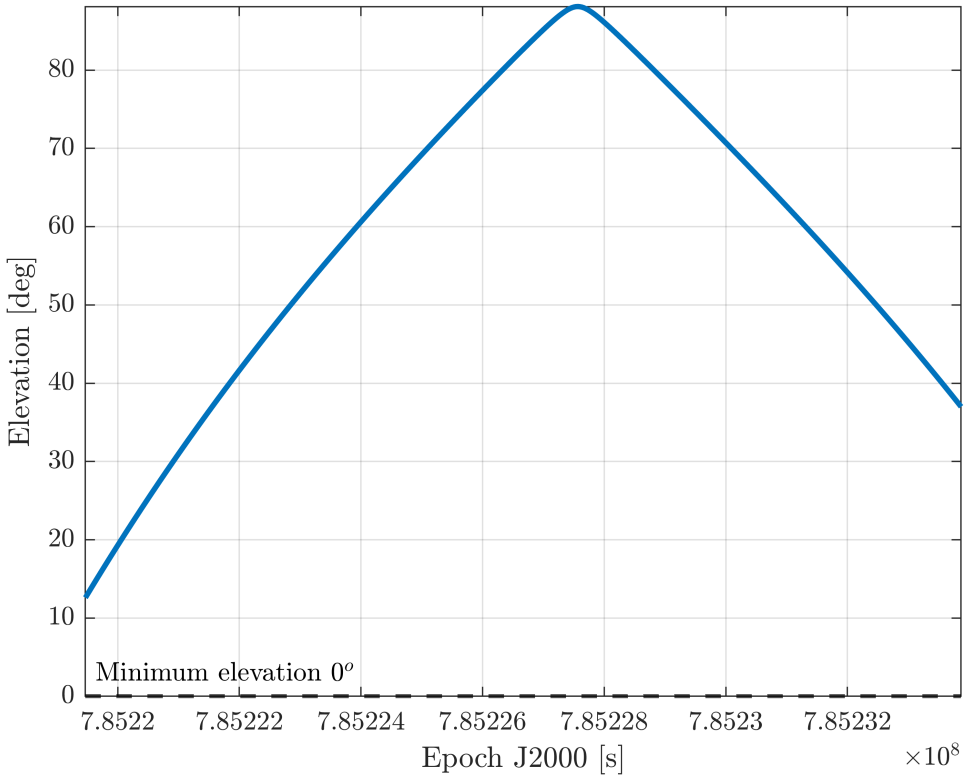


Figure 15: Orbiter estimated elevation (@Moonlander Topocentric reference)

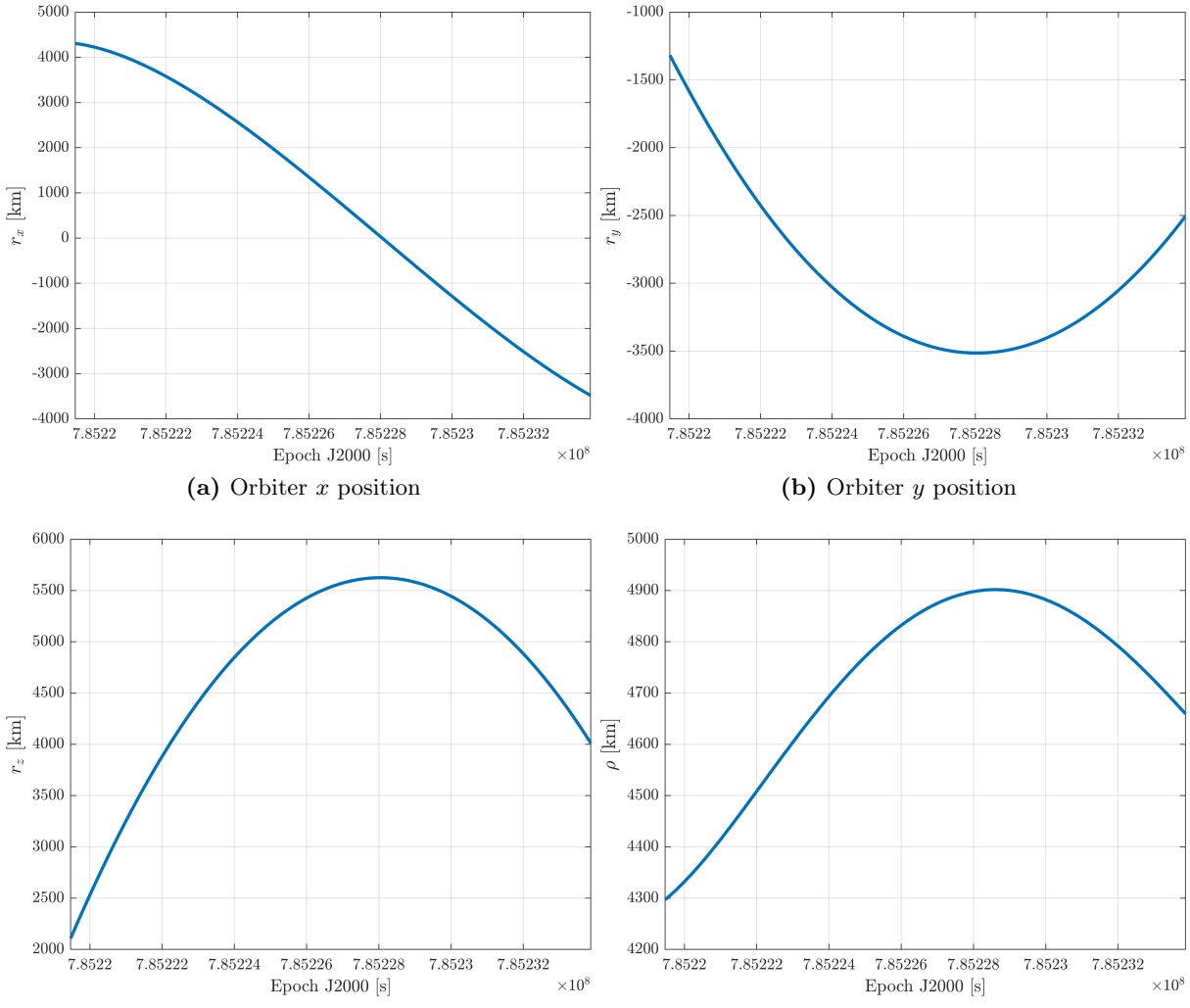
From Figure 15 is possible to observe that the elevation at each time instant is above the minimum one of the lander, so the orbiter and the lander are relative visibility for the entire time interval.

3.2 Measurements

The simulation of measurements has been carried out with the following procedure:

- propagation of the initial state of the orbiter, with a Keplerian propagator; then for each state in the time interval:
  - computation of the relative position between orbiter and lander;
  - extraction of the rotation matrix using `cspice_sxform` from Moon-Centered inertial frame to Moonlander Topocentric reference frame;
  - rotation of the relative position into lander fixed frame;
  - computation of the range between orbiter and lander;
- add noise modeled as Gaussian, to the range and the position of the orbiter, through `mvnrnd`.

The measurements are depicted in Figure 16.



(a) Orbiter x position

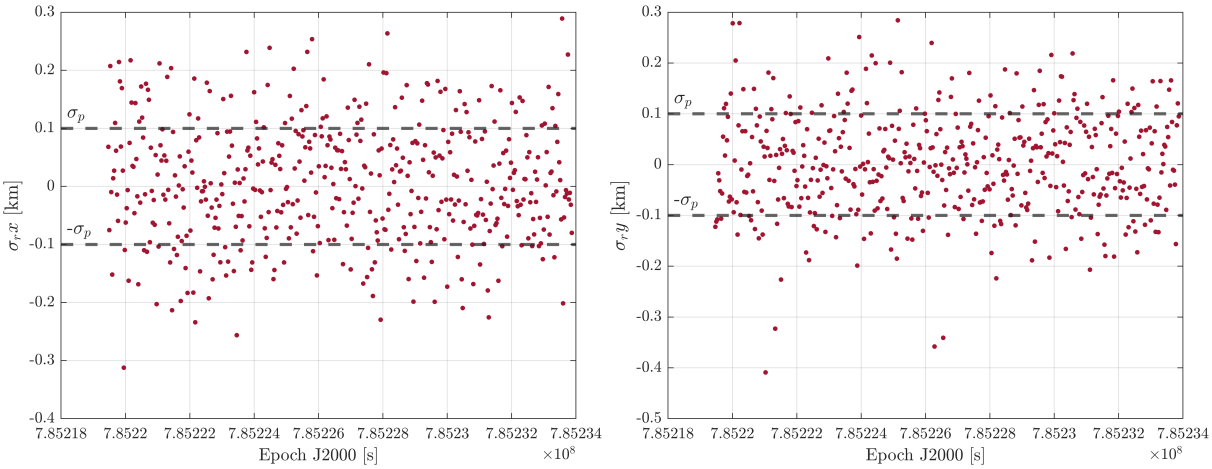
(b) Orbiter y position

(c) Orbiter z position

(d) Range

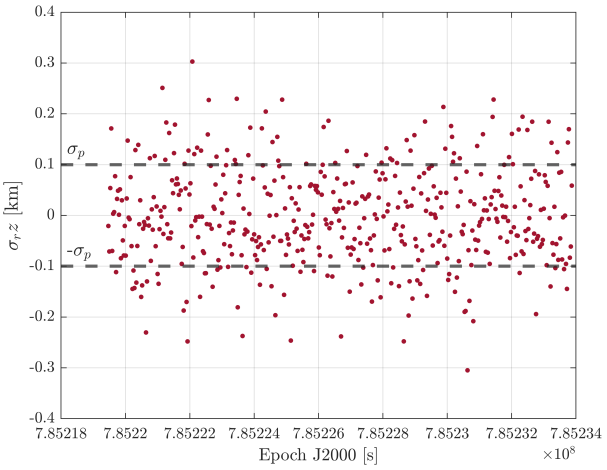
Figure 16: Orbiter measurements(@Moon Centered inertial reference frame)

The Figure 17 shows that the errors added to the position component and to the range are in the desired boundaries considering that the errors have been modeled as a Gaussian distribution, it is possible to observe that the majority of the errors are in bounded within $\pm\sigma_p$.

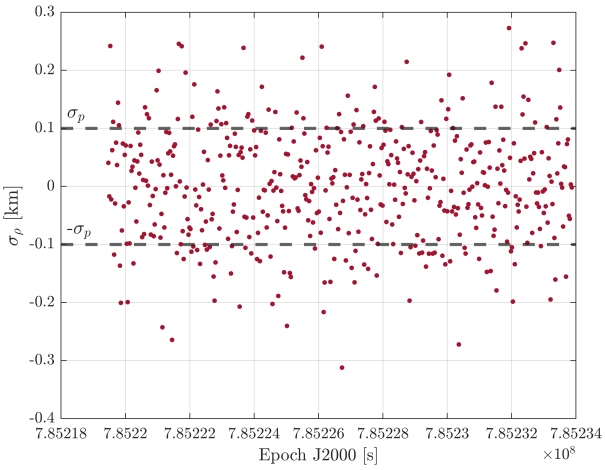


(a) Orbiter x position

(b) Orbiter y position



(c) Orbiter z position



(d) Range

Figure 17: Noise level of the measurements

3.3 Lunar orbiter absolute state

Now, to ensure an accurate estimation of covariance and mean state of the orbiter, the application of a filter becomes imperative. The selected filter for this task is the Unscented Kalman Filter (UKF).
To initialize the filter is necessary to know the mean value $\hat{\mathbf{x}}^+_{\mathbf{k-1}}$ and the covariance matrix $\mathbf{P}^+_{\mathbf{k-1}}$, these values correspond to the first epoch within the visibility window. In this scenario, $\hat{\mathbf{x}}^+_{\mathbf{k-1}}$ is equal to the initial state $\mathbf{x}_0$ perturbed through the function `mvnrnd` with covariance matrix $\mathbf{P}_0$ reported in, while $\mathbf{P}^+_{\mathbf{k-1}}$ is equal to $\mathbf{P}_0$ associated to position and velocity.
The algorithm of the UKF for a generic step from t_{k-1} to t_k, given $\hat{\mathbf{x}}^+_{\mathbf{k-1}}$, $\mathbf{P}^+_{k-1}$, and the measurement at t_k, $\mathbf{y}_k$, $\mathbf{R}_k$, works as follows:

- creation of the sigma points $\boldsymbol{\chi}_{i,k-1}$ and the weights using Equation 4, Equation 5, Equation 6.
- performing the prediction step
  - propagation of sigma points to t_k to obtain $\boldsymbol{\chi}_{i,k}$

- compute a prior mean

$$\hat{\mathbf{x}}_{\mathbf{k}}^{-} = \sum_{i=0}^{2N} W_i^{(m)} \boldsymbol{\chi}_{i,k} \tag{16}$$

- computation of a prior covariance;

$$\mathbf{P}_k^{-} = \sum_{i=0}^{2N} W_i^{(c)} \left(\boldsymbol{\chi}_{i,k} - \hat{\mathbf{x}}_{\mathbf{k}}^{-}\right) \left(\boldsymbol{\chi}_{i,k} - \hat{\mathbf{x}}_{\mathbf{k}}^{-}\right)^T \tag{17}$$

- computation of sigma point in measurements space, by applying the measurement function $\boldsymbol{\Upsilon}_{i,k} = \mathbf{h}(\boldsymbol{\chi}_{i,k})$
- computation of the mean of measurements;

$$\hat{\mathbf{y}}_{\mathbf{k}} = \sum_{i=0}^{2N} W_i^{(m)} \boldsymbol{\Upsilon}_{i,k} \tag{18}$$

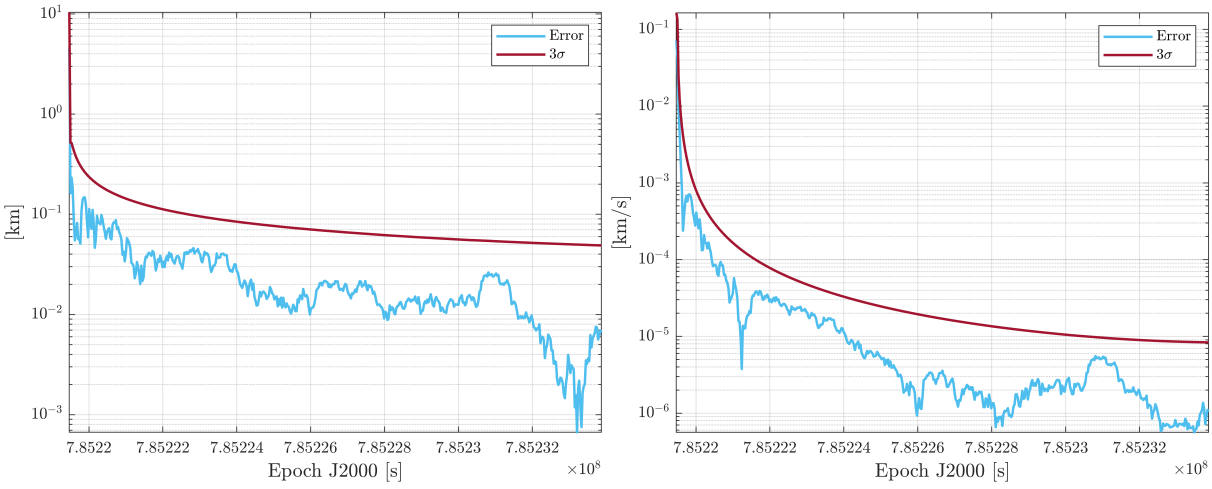
- Performing of updating step;
  - computation of measurement covariance and cross-covariance with state;

$$\mathbf{P}_{yy,k} = \sum_{i=0}^{2N} W_i^{(c)} \left(\boldsymbol{\Upsilon}_{i,k} - \hat{\mathbf{y}}_{\mathbf{k}}\right) \left(\boldsymbol{\Upsilon}_{i,k} - \hat{\mathbf{y}}_{\mathbf{k}}\right)^T + \mathbf{R}_k \tag{19}$$

$$\mathbf{P}_{xy,k} = \sum_{i=0}^{2N} W_i^{(c)} \left(\boldsymbol{\chi}_{i,k} - \hat{\mathbf{x}}_{\mathbf{k}}^{-}\right) \left(\boldsymbol{\Upsilon}_{i,k} - \hat{\mathbf{y}}_{\mathbf{k}}\right)^T \tag{20}$$

  - computation of Kalman gain and updating the state to retrieve a-posteriori mean and covariance;

$$\begin{cases} \mathbf{K}_k = \mathbf{P}_{xy.k} \mathbf{P}_{yy,k}^{-1} \\ \hat{\mathbf{x}}_{\mathbf{k}}^{+} = \hat{\mathbf{x}}_{\mathbf{k}}^{-} + \mathbf{K}_k \left(\mathbf{y}_k - \hat{\mathbf{y}}_{\mathbf{k}}\right) \\ \mathbf{P}_k^{+} = \mathbf{P}_k^{-} - \mathbf{K}_k \mathbf{P}_{yy,k} \mathbf{K}_k^T \end{cases} \tag{21}$$



(a) Position **(b)** Velocity

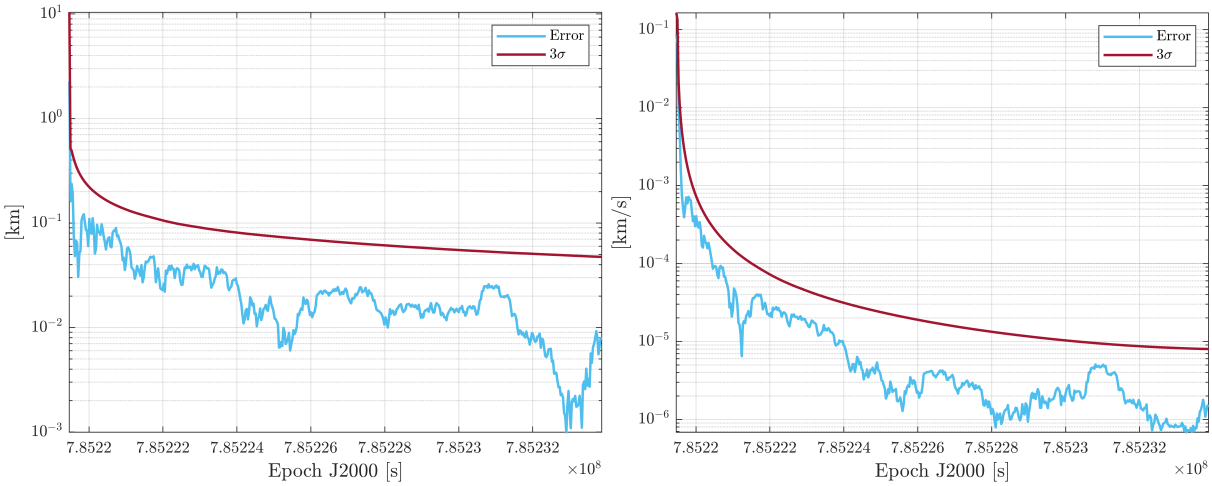
Figure 18: Estimated error with 3σ of estimated covariance of the orbiter absolute state)

The measurements are filtered using the UKF, and Figure 18 illustrates both the 3σ values and the error, calculated as the difference between the nominal state and the measured one. The outcomes show that the filter performs effectively, in fact, both position and velocity errors remain below the 3σ limit, which has a decreasing trend, this indicates an improvement in measurement precision as the algorithm operates.

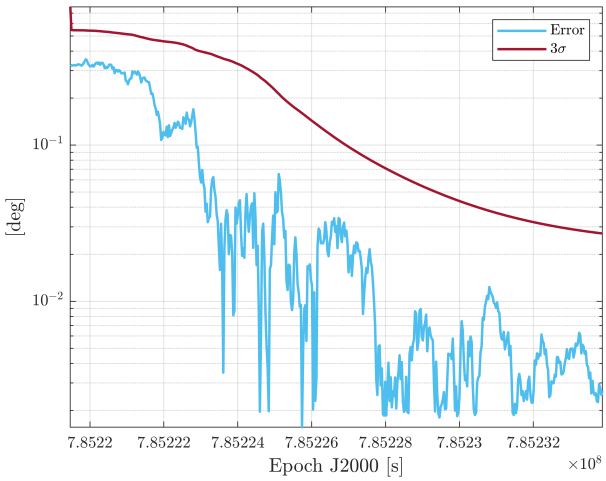
3.4 Lunar lander coordinates

To narrow down the uncertainty on the state of the orbiter and the lander coordinates, another UKF has been adopted, whose algorithm is the same explained in subsection 3.3, with some small modifications:

- $\hat{\mathbf{x}}^{+}_{\mathbf{k-1}}$ is generated with the same procedure explained in subsection 3.3, but the state is composed by the state of the orbiter and by the lander coordinates, for a total of 8 components;
- $\mathbf{P}^{+}_{k-1}$ is equal to $\mathbf{P}_0$;
- the range measurements have been used. In the UFK the range has been computed as the norm of the difference between the rotated state of the propagated sigma points, of the orbiter state, from Moon centered inertial frame to Moon centered fixed frame, and the converted sigma points, of lander coordinates, converted from geodetic to rectangular.



(a) Position **(b)** Velocity



(c) Velocity

Figure 19: Estimated error with 3σ of the estimated covariance of the orbiter absolute state, and the estimated lander coordinates

Figure 19 illustrates the 3σ values and the error, The outcomes show that, also in this case the filter performs effectively, in fact, position, velocity, and coordinates, errors remain below the 3σ limit.